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EVALUATION OF THE EAST LAGOON OREBODY,
MT. GUNSON, SOUTH AUSTRALIA
APRIL 1970

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Attachments (In pockets at back)

Contour Map of Copper Mineralisation
Isopach of Mineralisation
Contours on Top of + 0.4% Total Cu.
Contours on Bottom of + 0.4% Total Cu.
Contours on Top of Red Quartzite
Ore Reserve Blocks

Supplement (Separate cover)

East Lagoon Cross Sections

CONCLUSIONS AND RECOMMENDATIONS

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The ore reserves of the East Lagoon Orebody at Mt. Gunson, as of April, 1970, are calculated to be as follows :-

	<u>Short Tons</u>	<u>% Cu.</u>
Measured Ore	203,000	1.43
Indicated Ore	46,000	1.26
Inferred Ore	26,000	1.05

Continuity of ore grade and thickness for the central part of the orebody has been substantiated. The lateral limits of mineralisation have not been fully determined; only the southern limit has been reasonably established.

It is recommended that drilling be resumed as soon as possible so that the rest of the orebody may be delineated well in advance of mining operations.

	<u>ore</u>	<u>Contained Copper</u>
Total E. Lagoon	275,000 st	3,755 st
Mined to end Oct.	130,832 mt	* 913 mt

* Cu in Conc.

INTRODUCTION

In December, 1969, it was decided that the mining at Mount Gunson, scheduled to start in late April, 1970, would commence with the development of the East Lagoon Orebody.

The East Lagoon Orebody was selected because it appeared to contain the highest grade sulfide mineralisation even though the potential tonnage of ore was not as great as other deposits in the area. Equally important in the decision was the fact that drilling had suggested greater continuity of mineralisation in the East Lagoon deposit than in any of the other mineralised zones.

PREVIOUS DRILLING AND ESTIMATED RESERVES

The first drilling in the East Lagoon area was carried out in late 1967. Significant copper values were encountered in 12 percussion drill holes located on 200' x 100' centers. Eight of these percussion holes indicated a moderate size zone of potential ore.

In March, 1968, eight diamond drill holes were located immediately adjacent to the eight percussion holes. Only a fair degree of correlation was found to exist between some of the assays in the diamond drill holes and the adjacent percussion holes.

On the basis of the eight percussion drill holes, it was calculated that the ore reserves in the East Lagoon deposit were as follows :-

<u>Ore</u>	<u>Tonnage</u>	<u>Grade</u>	<u>Silver</u>
Measured	111,000T	1.75% Cu.	0.80 oz. Ag/T
Indicated	<u>43,000T</u>	<u>1.75% Cu.</u>	<u>0.80 oz. Ag/T</u>
<u>Total</u>	<u>154,000T</u>	<u>1.75% Cu.</u>	<u>0.80 oz. Ag/T</u>

(Review of Project Feasibility for 1,000 T.P.D. Mine and Mill, Mount Gunson Copper Project, South Australia by Wright Engineering Limited, Vancouver, Canada, September, 1968)

OBJECTIVES OF NEW DRILLING PROGRAMME

In order to prepare the East Lagoon Orebody for mining it was necessary to obtain the following pertinent information :-

1. The degree of continuity of mineralisation between the holes drilled in previous drilling programmes.
2. The areas and lateral limits of economic mineralisation.
3. More accurate ore reserve estimates.
4. Depths to lithologic units that would effect mining techniques.

To obtain this information a detailed geophysical survey and closely spaced drilling were scheduled to be carried out over the East Lagoon deposit.

GEOPHYSICS

The results of the detailed geophysical survey (induced polarisation) were used to assist in the location of drill holes.

A detailed report on the geophysical survey with a discussion of the results has been prepared under separate cover by Mr. A. Larson of Pacminex.

DRILLINGPROGRAMME

The drilling programme consisted of four continuous phases :-

1. Fill-in drilling between drill holes until the area of known mineralisation was covered by a 100' x 100' grid.
2. Drilling outward along each drill section at 50' centers until the lateral limits of mineralisation were determined.
3. Drilling at least two waste holes on each drill section beyond the limits of mineralisation.
4. Testing areas around the four holes encountering mineralisation outside the main mineralised zone and delineating if potential economic mineralisation was indicated.

DRILLING METHOD

Percussion drilling was chosen for the drilling programme for the following reasons :-

(a) Speed

Results had to be obtained as soon as possible so that the compiled data would be available for planning of the mining operations. Percussion drilling is a rapid means of sampling.

(b) Availability

A percussion drill rig was available at the site.

(c) Cost

Percussion drilling is relatively inexpensive, the cost being approximately \$1.00/ft.

SAMPLING AND LOGGING

During the drilling of each hole the zones of potential mineralisation were sampled on two foot intervals. The samples were simultaneously logged to obtain data on depths to lithologic units, percentage and species of minerals encountered and estimated grades of copper.

The samples were split to convenient size; wet samples being dried before splitting. The samples were subsequently bagged in numbered envelopes for shipment to the assay laboratory.

ASSAYING

The samples were sent to the C.S.R. Central Laboratory for preparation and copper analysis by atomic absorption. Results from the laboratory were reported as percent total copper.

Each drill hole showing significant copper values was subsequently composited. The composited samples were analysed by atomic absorption for silver. Results were reported in parts per million which were converted to ounces (Troy) per short ton.

Five composited drill holes have been selected for complete analysis. These samples will be analysed as soon as the new x-ray system at the Central Laboratory is operational.

DRILLING PROBLEMS AND SAMPLE RELIABILITY

Throughout the drilling programme various sampling problems have been apparent.

As long as the rock being drilled is dry the drilling is rapid and there is good sample reliability. If the hole is drilled quickly it can often be completed before significant water enters the hole. As soon as the samples become wet, however, sample reliability drops and drilling is impeded. The drilling of a new hole and casing through the water producing zone is often necessary to achieve even fair sample reliability.

The most troublesome water producing zone in the East Lagoon is in the lower part of the lithic sandstone immediately on top of the disconformity with the red quartzite. Sample reliability in this zone was often poor and it is probable that the better copper values expected at this interface were often inadequately sampled.

The percussion drilling has occasionally led to the salting of copper values. After the drilling of a wet hole some copper values may remain in the sampling system and contaminate the samples from the next hole. Adequate cleaning of the sampling system is necessary to avoid this type of salting. Salting may also occur after the drill has penetrated a mineralised section with copper values from above contaminating samples from below.

Recognition of salting was not always possible. In all cases where salting was detected the erroneous assays have been disregarded.

Upon occasion the sampling of a hole did not begin soon enough and sometimes drilling or sampling was stopped too soon. Therefore, there are holes in which the top or bottom of ore grade mineralisation has not been determined. In most of these cases, rather than drill new holes, the existing data was utilized, especially if continuity of mineralisation had been established.

SUMMARY OF DRILLING RESULTS

GENERAL

The drilling programme was scheduled to commence in early January, 1970. However, due to a lack of equipment, actual production drilling did not start until early February. Drilling progress was slow with poor sample reliability and it was not until an experienced driller had been obtained that the drilling progressed smoothly and rapidly.

In early March the drilling was terminated before the objectives of the programme could be achieved. This was due to the departure of the experienced driller who had previous commitments elsewhere.

The final drill hole co-ordinates and assay results were completed by late March.

A total of 60 locations were sampled. At many locations, however, more than one hole was necessary before adequate sampling could be achieved. In some cases the data from two or more holes were combined so as to form an idealized hole considered as representative of the area. The data is thus presented as 60 drill holes.

Of the 60 drill holes 40 intersected significant copper mineralisation and 20 were drilled in waste.

As mentioned above, the drilling was terminated before the objectives of the programme were fully realized. Fill-in drilling to a 100' x 100' grid was completed. Internal continuity of ore grade and thickness for the center of the main zone of mineralisation was substantiated.

The lateral limits of the mineralisation have not been determined. Step-out drilling along drill sections at 50' centers continued to encounter significant copper mineralisation.

Mineralisation trends on the west, east, and northeast sides are still open and considerable reserves may yet be found in these directions.

To the north the thickness of mineralisation is decreasing and some holes have been drilled in waste. However, more drilling is necessary to determine if the economic limits have been reached.

Only on the south side of the deposit have sufficient holes been drilled to reasonably establish a lateral limit to mineralisation. Here, on most sections, there are at least two waste holes outside of the ore zone.

COMPILATION OF DATA

Three drilling programmes of various degrees of reliability are represented on both plans and sections. The programmes are as follows :-

<u>Hole Prefix</u>	<u>Example</u>	<u>Date Drilled</u>	<u>Comments</u>
P	P184-137	Sept.-Oct. 1968	Percussion drilling, 5' sample intervals.
D	D184-137	March 1968?	Diamond drilling, 2.5' sample intervals.
3	3-1-27P	Feb.-March 1968 70	Percussion drilling, 2' sample intervals.

In the last example listed above (3-1-27P) the number 3 refers to the project (Mt. Gunson), number 1 refers to the area (East Lagoon), 27 refers to the hole number and P refers to the type of drilling (percussion).

The initial drilling results in the East Lagoon (Sept.-Oct. 1968) are considered to be the least reliable and little consideration has been given to the assay data. The diamond drilling (March 1968), carried out to check the earlier percussion drilling, is considered to have given the best results in the area. The latest percussion drilling (February-March, 1970) has been the most extensive programme and the results are considered to have fair reliability.

The diamond drilling data has been used in conjunction with the latest percussion drilling programme to evaluate the East Lagoon deposit.

MINERALISATION GRADES

A mining cut-off grade of 0.4% total copper has been used in all calculations. The mineralisation grades utilized are as follows :-

Waste	Less than 0.4% total copper
Low grade sulphide ore	0.4% - 1.00% T. Cu.
Medium grade sulphide ore	1.00 - 1.50% T. Cu.
High grade sulphide ore	More than 1.50% T. Cu.

COMPOSITING OF ASSAY VALUES

The average thickness of mineralisation (11.5') is thin and can be effectively mined in one level. For this reason, copper values for each drill hole have been composited so that one value may be assigned to each drill hole.

PLANS

The following plans of the East Lagoon Orebody have been prepared :-

- (a) Contour Map of Copper Mineralisation.
- (b) Isopach of Mineralisation (>0.4% Total Cu.).
- (c) Contours on Top of + 0.4% Total Cu.
- (d) Contours on Bottom of + 0.4% Total Cu.
- (e) Contours on Top of Red Quartzite.
- (f) Ore Reserve Blocks.

The plans are mostly self-explanatory, however, it should be noted that the mineralisation is confined to a sedimentary sequence and is blanket-like in configuration. For these reasons it is considered that the mineralisation lenses out laterally rather than terminating abruptly.

No attempt has been made to interpret the geologic significance of the plans. As noted previously, there are occasional holes where either the top or bottom of mineralisation was not adequately determined. Correct data from these holes would naturally change some of the plans. However, two plans require some further comment.

1. Contour Maps of Copper Mineralisation

This plan emphasizes the high grade core of the orebody with a distinct gradation laterally into low grade mineralisation. It also suggests that the copper mineralisation may be surrounded by a pyritic halo (mineral zoning?).

2. Isopach of Mineralisation

The isopach map shows a rather irregular pattern due to the above mentioned incomplete sampling. However, the blanket-like nature of the deposit is well demonstrated. This map, when used in conjunction with the Contour Map of Copper Mineralisation, indicates the better mining sectors of the orebody.

SECTIONS

A total of 24 sections have been prepared and are presented separately in a booklet accompanying this report.

The sections, nine north-south sections at 100' intervals and fifteen east-west sections at 50' intervals, illustrate the blanket-like configuration of the deposit. They also show how most of the ore occurs within the red quartzite.

Again, incomplete sampling and logging has left gaps in the geological picture. Ore intercepts on the sections are occasionally incomplete. Also, the silicified lithic sandstone has not always been noted during logging. Further work may prove that this unit has greater continuity than depicted on the sections.

ORE RESERVE ESTIMATES

The ore reserves have been divided into three categories based upon the degree of assurance of existence. The classes are as follows :-

<u>Measured Ore</u>	Ore delineated by detailed drilling so that size, shape, and mineral content are well established.
<u>Indicated Ore</u>	Ore lacking in sufficient drilling so that size, shape, and mineral content are uncertain.
<u>Inferred Ore</u>	No drilling; estimates based upon assumed continuity for which there is geologic evidence.

ASSUMPTIONS

1. Sulphide cut-off grade = 0.4% total copper.
2. Specific gravity of ore = 2.60.
3. Mining can be effected from one level, hence no significant lateral dilution of ore.
4. Mining control will admit no more than one foot above and one foot below mineralisation as dilution material.

METHOD

The accompanying Plan of Ore Reserve Blocks shows the areas assigned to each ore category.

Blocks 1-55 were calculated by the triangular method. The peripheral blocks 56-83 were calculated by the rectangular method. The two results were combined to give the measured ore reserves.

The indicated ore, blocks 84-90 and the inferred ore, blocks 91-93 were calculated by the rectangular method.

Blocks 94 and 95 are isolated from the main zone of mineralisation and each contain one percussion drill hole that encountered significant copper mineralisation. Both areas require drilling to assess their potential.

Block 96 is located immediately adjacent to the unclosed eastern end of the orebody. Mineralisation appears to be thinning in that direction, however, the block should be tested to ensure that no ore is missed.

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ORE RESERVE CALCULATION SUMMARY - APRIL 1970

UNDILUTED ORE

DILUTION MATERIAL

<u>Ore Category</u>	<u>Tons (Short)</u>	<u>Avg. Grade % Total Cu</u>	<u>Over - burden</u>	<u>Stripping Ratio Overburden/ Ore</u>	<u>Tons (Short)</u>	<u>Avg. Grade % Total Cu</u>
Measured	203,000	1.43	422,000	2.1:1	35,000	0.17
Indicated	<u>46,000</u>	<u>1.26</u>	<u>150,000</u>	<u>3.3:1</u>		
<u>Sub-Total</u>	249,000	1.39	572,000	2.3:1		
Inferred	26,000	1.05	42,000	1.6:1		

SILVER

As previously mentioned, each drill hole with significant copper values was composited for silver analysis. One silver value has thus been assigned to each drill hole. The results of these analyses, in Troy ounces/ton, are listed in Appendix 2.

Insufficient data was available for the preparation of a silver grade contour map. However, it is apparent that the better silver values are generally associated with the medium and high grade copper assays.

It is also impractical to assign an average silver grade to the orebody. However, a compositing of the drill holes (footage x grade) gives an average value of 0.28 Troy ounces/ton for the orebody. This figure should be considered very tenuous at best.

MEASURED ORECENTER

<u>Triangle</u>	<u>Area Ft.2</u>	<u>H1</u>	<u>G1</u>	<u>H2</u>	<u>Ore G2</u>	<u>H3</u>	<u>G3</u>	<u>Overburden</u>		
								<u>H1</u>	<u>H2</u>	<u>H3</u>
1	2,111	2.5	4.48	1.1	3.30	8	0.64	22.5	35	44
2	1,488	2.5	4.48	2.0	3.63	11.0	3.30	22.5	20	35
3	2,544	2	3.63	11.0	3.30	18.0	0.89	20	35	46
4	2,401	11	3.30	18	0.89	15	1.21	35	46	35
5	2,499	2	3.63	18	0.89	26	0.93	20	46	24
6	2,303	26	0.93	18	0.89	5	1.21	24	46	35
7	1,519	26	0.93	5	1.21	16	1.52	24	35	28
8	1,887	5	1.21	16	1.52	12	0.86	35	28	28
9	1,196	5	1.21	12	0.86	12	0.70	35	28	30
10	1,000	16	1.52	12	0.86	0	0	28	28	0
11	1,152	12	0.86	12	0.70	6	0.43	28	30	28
12	1,734	8	1.01	6	1.29	4	0.64	20	28	24
13	2,550	8	1.01	6	1.29	6	0.43	20	28	28
14	2,397	6	0.43	12.5	1.38	6	1.29	28	25	28
15	2,499	6	0.43	12.5	1.38	12	0.70	28	25	30
16	4,896	12.5	1.38	12	0.70	20	2.82	30	25	20
17	2,550	12	0.70	5	1.21	20	2.82	30	35	20
18	3,000	5	1.21	20	2.82	18	1.51	35	20	28
19	4,949	5	1.21	18	1.51	11	3.30	35	28	35
20	1,540	11	3.30	18	1.51	25	2.06	35	28	30
21	3,417	11	3.30	25	2.06	8	0.98	35	30	40
22	2,180	11	3.30	8	0.98	8	0.64	35	40	44
23	2,494	8	0.98	7	0.61	14	1.08	40	43	40
24	2,565	8	0.98	14	1.08	11	1.30	40	40	39
25	3,477	8	0.98	11	1.30	25.0	2.06	40	20	30
26	2,652	11	1.30	25	2.06	36	1.18	30	39	14
27	1,519	25	2.06	36	1.18	18	1.51	30	14	28
28	2,544	36	1.18	18	1.51	8	1.44	28	14	28
29	2,820	18	1.51	8	1.44	20	2.82	28	28	20

<u>Triangle</u>	<u>Area</u> <u>Ft.2</u>	<u>H1</u>	<u>G1</u>	<u>H2</u>	<u>Ore</u> <u>G2</u>	<u>H3</u>	<u>G3</u>	<u>H1</u>	<u>Overburden</u> <u>H2</u>	<u>H3</u>
30	4,690	20	2.82	8	1.44	16	1.62	20	28	16
31	4,800	20	2.82	16	1.62	12.5	1.38	20	16	25
32	2,280	16	1.62	12.5	1.38	6	1.29	25	16	28
33	3,600	16	1.62	6	1.29	8	1.32	16	28	18
34	3,180	16	1.62	8	1.32	14	1.44	16	18	18
35	832	8	1.32	14	1.44	14	1.12	18	18	20
36	4,526	14	1.44	16	1.62	7.5	0.73	16	18	17.5
37	4,355	16	1.62	13.5	1.67	7.5	0.73	16	14	17.5
38	4,355	16	1.62	13.5	1.67	12	2.64	16	14	17
39	5,000	16	1.62	12	2.64	8	1.44	16	17	28
40	2,600	8	1.44	12	2.64	36	1.18	28	17	14
41	3,672	12	2.64	36	1.18	4	2.96	14	17	24
42	2,626	36	1.18	4	2.96	11	1.30	14	24	39
43	3,264	11	1.30	4	2.96	4	1.34	39	24	30
44	2,376	11	1.30	4	1.34	14	1.08	39	30	40
45	2,337	14	1.08	4	1.34	7	0.61	40	30	43
46	4,160	4	2.96	12	2.64	14	0.71	24	17	30
47	5,850	12	2.64	14	0.71	20	0.63	17	30	12
48	5,175	12	2.64	20	0.63	13.5	1.67	17	12	14
49	5,054	13.5	1.67	20	0.63	7.5	0.73	14	127	17.5
50	4,686	7.5	0.73	20	0.63	2	0.60	12	17.5	22
51	2,618	7.5	0.73	2	0.60	4	1.05	17.5	22	22
52	5,390	7.5	0.73	5.0	0.65	4	1.05	17.5	17.5	22
53	4,028	14	0.71	20	0.63	18	0.51	30	12	14
54	1,815	14	0.71	18	0.51	4	1.05	30	14	12
55	2,750	14	0.71	4	1.05	4	3.12	30	12	14

Area 165,902 ft.²
 Volume 2,100,480 ft.³
 Specific Gravity 2.60
 1 Ton = 12.33 ft.³
 Tons 170,356
 Grade 1.50% Total Cu.

Area 165,902 ft.²
 Volume 4,047,086 ft.³
 Specific Gravity 2.60
 1 Ton = 12.33 ft.³
 Tons 328,231

MEASURED OREPERIPHERY

<u>Block</u>	<u>Area</u> <u>Ft.²</u>	<u>H Ft.</u>	<u>Ft.³ Ore</u>	<u>Tons</u>	<u>Grade</u>	<u>H Ft.</u>	<u>Overburden</u> <u>Ft.³</u>	<u>Tons</u>
56	1,350	5	6,750	568.7	0.70	41	55,350	4,489
57	604	5	3,020	254.4	2.50	23	13,892	1,127
58	981	2.5	2,452.5	206.6	4.30	22	21,582	1,750
59	1,745	4	6,980	588.0	3.20	20	34,900	2,830
60	2,050	14	28,700	2,417.9	0.90	22	45,100	3,658
61	1,605	23	36,915	3,109.9	0.93	24	38,520	3,124
62	1,665	12	19,980	1,683.2	1.50	28	46,620	3,781
63	1,330	8	10,640	896.4	0.70	28	37,240	3,020
64	2,650	5	13,250	1,116.3	0.50	27	71,550	5,803
65	2,590	5	12,950	1,091.0	0.90	22	56,980	4,621
66	810	5	4,050	341.2	0.65	23	18,630	1,511
67	692	5	3,460	291.5	0.90	25	17,300	1,403
68	1,809	4	7,236	609.6	1.00	25	45,225	3,668
69	1,059	6	6,354	535.3	1.30	21	22,239	1,804
70	2,460	10	24,600	2,072.5	1.10	19.5	47,970	3,891

<u>Block</u>	<u>Area</u> <u>Ft.²</u>	<u>H Ft.</u>	<u>Ore</u>			<u>Grade</u>	<u>Overburden</u>		
			<u>Ft.³</u>	<u>Tons</u>			<u>H. Ft.</u>	<u>Ft.³</u>	<u>Tons</u>
71	1,763	9	15,867	1,336.7		1.30	17.5	30,852.5	2,502
72	1,980	6	11,880	1,000.8		0.80	17.5	34,650	2,810
73	1,077	5	5,385	453.7		0.70	17.5	18,847.5	1,529
74	1,935	4	7,740	652.1		0.75	18	34,830	2,825
75	2,290	4	9,160	771.7		1.00	22	50,380	4,086
76	1,800	7	12,600	1,061.5		0.62	18	32,400	2,628
77	2,470	12	29,640	2,497.1		0.60	13	32,110	2,604
78	2,355	15	35,325	2,976.0		0.60	13	30,615	2,483
79	2,250	4	9,000	758.2		1.10	12	27,000	2,190
80	3,870	5	19,350	1,630.2		2.50	15	58,050	4,708
81	2,180	10	21,800	1,836.6		1.00	27	58,860	4,774
82	3,110	5	15,550	1,310.0		2.50	26	80,860	6,558
83	3,220	6	19,320	1,627.6		1.25	29	93,380	7,573

Area 53,600 ft.²
Volume 399,954 ft.³
Specific Gravity 2.60
1 Ton = 12.33 ft.³
Ton 32,437
Grade 1.06% Total Cu.

Area 53,600 ft.²
Volume 1,155,933 ft.³
Specific Gravity 2.60
1 Ton = 12.33 ft.³
Tons 93,750

Total Measure Ore Reserves (Undiluted)
Overburden over Measured Ore Reserves

= Center + Periphery = 202,789 tons at 1.43% Total Cu.
= Center + Periphery = 421,981 tons

Total Measured Ore Reserves (Diluted)

Assume 1 ft. of dilution material from above and below ore = 2 ft. dilution
Average ore intercept = 11.5 ft.
2 ft. of dilution in 11.5 ft. = 17.4% dilution
17.4% of 202,789T = 35,322 T dilution material
Average grade of dilution material = 0.17% Total Cu.

Total Measured Ore Reserves (Diluted) = 238,111 tons at 1.24% Total Cu.